SCHOOL OF COMPUTING TECHNOLOGY AND APPLIED SCIENCES

CCS1102 FUNDAMENTALS OF PROGRAMMING

FINAL SEMESTER EXAMINATION

TUESDAY 3 DECEMBER, 2024

14:00 - 17:00 HOURS

TIME ALLOWED : WRITING – THREE HOURS : READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

Question 1

a) Desk check the python code given below. **(10 Marks)**

assume the following:

```
input_salary  = 40000
input_credit_score  = 1000
input_years_employed  = 5
```

```
1.  MINIMUM_SALARY = 30000
2.  MINIMUM_CREDIT_SCORE = 650
3.  MINIMUM_YEARS_EMPLOYED = 2
4.  BASE_LOAN_AMOUNT = 50000
5.  SALARY_MULTIPLIER = 0.2
6.  CREDIT_SCORE_MULTIPLIER = 100
7.  EMPLOYMENT_MULTIPLIER = 500
8.
9.  input_salary = float(input("Enter the applicant's annual salary: "))
10. input_credit_score = int(input("Enter the applicant's credit score: "))
11. input_years_employed = int(input("Enter the applicant's years of employment: "))
12.
13. if input_salary < MINIMUM_SALARY:
14.     loan_eligibility = "Not Eligible: Insufficient salary"
15.     actual_loan_amount = 0
16. elif input_credit_score < MINIMUM_CREDIT_SCORE:
17.     loan_eligibility = "Not Eligible: Low credit score"
18.     actual_loan_amount = 0
19. elif input_years_employed < MINIMUM_YEARS_EMPLOYED:
20.     loan_eligibility = "Not Eligible: Insufficient years of employment"
21.     actual_loan_amount = 0
22. else:
23.     loan_eligibility = "Eligible"
24.     actual_loan_amount = BASE_LOAN_AMOUNT
25.     actual_loan_amount += input_salary * SALARY_MULTIPLIER
26.     actual_loan_amount += (input_credit_score - MINIMUM_CREDIT_SCORE) *
            CREDIT_SCORE_MULTIPLIER
27.
28.     actual_loan_amount += (input_years_employed - MINIMUM_YEARS_EMPLOYED) *
            EMPLOYMENT_MULTIPLIER
29.
30. print("Loan eligibility status: " + loan_eligibility)
31.
32. if loan_eligibility == "Eligible":
33.     print("Actual loan amount to give", actual_loan_amount)
```

b) Desk check the python *code* given below. **(10 Marks)**

Assume the following:

iteration 1
purchase = 200
choice = Y
iteration 2
purchase = 200
choice = Y
iteration 3
purchase = 200
choice = N

```
1.  REWARD_THRESHOLD = 500
2.  customers_processed = 0
3.
4.  print("Welcome to the Customer Loyalty Rewards System!")
5.
6.  while customers_processed < 5:
7.          total_spent = 0
8.          print("\nProcessing new customer...")
9.
10.         purchase = float(input("Enter the amount spent in the latest purchase: "))
11.         total_spent += purchase
12.         print("Current total spending",total_spent)
13.
14.         if total_spent >= REWARD_THRESHOLD:
15.                 print("Congratulations! The customer qualifies for a loyalty reward.")
16.                 customers_processed += 1
17.         else:
18.                 print("Customer has not reached the reward threshold.")
19.
20.         choice = input("Would you like to process another customer? (Y/N): ")
21.
22.         if choice != "Y":
23.                 print("Exiting the loyalty rewards system.")
24.                 break
25.
26. print("Total customers processed: ", customers_processed)
27. print("Loyalty rewards processing completed.")
```

c) Explain five advantages of using pseudocode before writing the actual code in a programming language. **(10 Marks)**

d) With the aid of appropriate examples, compare and contrast high-level programming *languages* and low-level languages. **(10 Marks)**

[Total: 40 Marks]

SECTION B: Attempt any THREE questions in this section

Question 2

a) Below is a table of variable declarations. State whether the declarations are true or false in the validity column. **(10 Marks)**

Variable	Validity
average_score	
order_date	
min/value	
user_name	
is@valid	
order.date	
employee id	
average score	
customer_feedback	
max#value	
total_price	
total-price	
customer-feedback	
product$list	
employee_id	
min_value	
is_valid	
max_value	
product_list	
123user	

b) In a retail store, a bulk purchase discount is applied based on the number of items bought. If a customer buys more than 50 items, they get a 10% discount on the total cost. If they buy 50 or fewer items, no discount is applied.

Write a Python program that takes the number of items purchased and the price per item as inputs from the user. The program should calculate the discount (if applicable) and display t he total cost after applying the discount. **(10 Marks)**

[Total: 20 Marks]

Question 3

a) A bank categorizes loan applicants based on their credit score. Write a Python program that takes a credit score as input and classifies the applicant into the following categories:

"Excellent" for scores >= 750
"Good" for scores >= 700
"Fair" for scores >= 650
"Poor" for scores < 650
The program should display the credit score and the corresponding classification.

(10 Marks)

b) Write down the correct letter corresponding to the answer. **(10 Marks)**

1. What is the correct way to declare a constant variable in Python?
   A) CONSTANT = 100
   B) const CONSTANT = 100
   C) constant = 100
   D) let constant = 100

2. Which Python keyword is used to exit a loop prematurely?
   A) exit
   B) break
   C) stop
   D) terminate

3. What happens if indentation is not consistent in a Python script?
   A) The script runs with a warning.
   B) The script skips the incorrectly indented block.
   C) The script throws an IndentationError.
   D) The script corrects the indentation automatically.

4. Which of the following operators is used for floor division in Python?
   A) /
   B) //
   C) %
   D) \

5. Which method is called to represent an object as a string in Python?
   A) __repr__()
   B) __str__()
   C) __call__()
   D) __init__()

6. What will the value of y be after executing the following code?

```
y = 10
y -= 4 + 2
```

A) 4
B) 8
C) 6
D) 3

7. Which Python data structure is mutable and allows duplicate values?
A) tuple
B) list
C) set
D) string

8. Which Python keyword is used to check multiple conditions sequentially?
A) elif
B) switch
C) elseif
D) case

9. What is the result of the expression 15 % 4 in Python?
A) 3.0
B) 3
C) 3.75
D) 11

10. What does the @staticmethod decorator indicate in Python?
A) The method doesn't need access to the class or instance.
B) The method is abstract and must be overridden.
C) The method is private to the class.
D) The method modifies the class itself.

[Total: 20 Marks]

Question 4

a) Critically study the code below and show what the output will be. **(10 Marks)**

```
1. class Person:
2.     def __init__(self, name, age):
3.         self.name = name
4.         self.age = age
5.
6.     def display_info(self):
7.         print(f"My name is {self.name} and I am {self.age} years old.")
8.
9.     def is_adult(self):
10.        if self.age >= 18:
11.            print(f"{self.name} is an adult.")
12.        else:
13.            print(f"{self.name} is not an adult.")
14.
15. person1 = Person("Alice", 22)
16. person1.display_info()
```

```
17. person1.is_adult()
18.
19. person2 = Person("Bob", 15)
20. person2.display_info()
21. person2.is_adult()
```

b) Critically study the pythone code below, detect syntax errors and correct them. **(10 Marks)**

```
1.  value1 = 10
2.  value2 = 5
3.
4.  if value1 > value2
5.      difference = value1 - value2
6.      print("The difference is: " difference)
7.
8.  if value1 % 2 = 0:
9.      print(f"{value1} is an even number")
10. else
11.     print(f"{value1} is an odd number")
12.
13. total = value1 + value2
14. print("The total is: " total)
```

[Total: 20 Marks]

Question 5

a) Write a python function to calculate the perimeter of a rectangle given the formular:

$P=2\times(l+w)$ *where*

P: Perimeter

l: Length

w: Width

(10 Marks)

b) Study the program given below critically and show the final output. **(10 Marks)**

```
1. elements = [10, "apple", 42, "banana", 8, "cherry", 56, "date", 23, 5]
2. first_element = elements[0]
3. second_element = elements[1]
4.
5. elements[4] = "orange"
6.
7. elements.append(77)
8.
9. elements.remove(42)
```

```
10.
11. elements.insert(3, "kiwi")
12.
13. list_length = len(elements)
14.
15.
16. elements.reverse()
17.
18. subset = elements[2:6]
19.
20. if "banana" in elements:
21.     print("banana is in the list")
22.
23. for element in elements:
24.     print(element)
```

[Total: 20 Marks]

END OF EXAMINATION